



Implications of changes in WHO haemoglobin elevation adjustment guidelines on global, regional, and national anaemia burden, 1990–2023: a population-based modelling study

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Summary

Background Adjustment of haemoglobin concentrations for elevation is essential for anaemia assessments of both individuals and populations. Whereas the previous WHO anaemia definitions released in 2001 used an elevation adjustment derived from small sample sizes with limited representativeness, the 2024 WHO global anaemia definitions reflect empirical assessments across several geographies and settings derived from population-representative surveys. We aimed to assess the effect of these changes on estimates of global anaemia burden.

Methods In this population-based modelling study, using an identical set of input data from population-based surveys, we created two datasets in parallel, where each input haemoglobin concentration was adjusted using the WHO 2001 and WHO 2024 elevation adjustment methods. We estimated mild, moderate, and severe anaemia prevalence by age and sex for 204 countries and territories from 1990 to 2023 for each dataset using spatiotemporal Gaussian process regression and ensemble distribution modelling. We then analysed the overall differences in anaemia prevalence and years lived with disability to assess the effect of the WHO 2024 elevation adjustment method on the global landscape of anaemia. Uncertainty was propagated in each step of the modelling process by sampling draws of the posterior distribution of each estimated quantity.

Findings Global anaemia prevalence for all ages and both sexes in 2023 was estimated to be 26·4% (95% uncertainty interval [UI] 22·6–31·9) using the WHO 2024 elevation adjustment method, which was 2·4 percentage points (2·2–2·7) higher than the total anaemia prevalence estimated using the WHO 2001 method (24·0% [20·6–29·3]). This increase equates to 198 million (196–199) newly enumerated cases, increasing the rank of anaemia from the third to the second largest cause of disability globally. The largest absolute increases in anaemia prevalence were in locations between 500 m and 2000 m in elevation, which include countries in eastern sub-Saharan Africa, central Asia, central America, and South America.

Interpretation Understanding the effect of anaemia requires unbiased and comparable estimates of anaemia burden. To our knowledge, we produced the first set of global estimates of anaemia burden by location, year, age, and sex using the WHO 2024 elevation adjustment method and compared them with estimates using the previous method. Policy makers should consider this modification when designing interventions to manage and prevent anaemia, particularly in regions most affected by changes in elevation adjustment.

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Introduction

Anaemia affected nearly 2 billion people and was the third leading cause of morbidity worldwide in 2021.¹ Despite global efforts, its prevalence has remained high for decades, prompting renewed calls for action.² Accurate anaemia prevalence estimates are vital for monitoring progress towards international reduction targets, especially in the light of the WHO 2024 guideline updates.³

Anaemia is present when an individual has insufficient red blood cell oxygen delivery to meet the body's needs.⁴

Anaemia is diagnosed by proxy as having a haemoglobin concentration below a defined age-specific, sex-specific, and pregnancy-specific threshold.³ Anaemia is linked to adverse health outcomes, particularly among children, adolescent girls, women, and older adults (>65 years).^{5,6} Several factors—most notably elevation and chronic smoke exposure—cause haemoglobin increases without increased oxygen delivery via hypoxia-induced stimulation of erythropoietin and production of red blood cells.^{7,8}

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See Online for appendix

For the WHO Vitamin and Mineral Nutrition Information System see <https://www.who.int/teams/nutrition-and-food-safety/databases/vitamin-and-mineral-nutrition-information-system>

Research in context

Evidence before this study

Recent evidence from the Biomarkers Reflecting Inflammation and Nutritional Determinants of Anemia (BRINDA) project found that the WHO 2001 guidelines, which are based on data lacking geographical and demographic diversity, under-adjust haemoglobin concentrations at lower elevations and over-adjust them at higher elevations, leading to systematic bias in estimates of global anaemia burden. Recognising these limitations, WHO updated its haemoglobin elevation adjustment recommendations in 2024 to align with evidence from BRINDA's more diverse cohorts. The Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) comprehensively quantifies health loss across space and time, including routine estimation of anaemia burden, but up until now has not reflected the updated elevation adjustment method for haemoglobin concentration. We conducted a systematic literature search in PubMed using the search terms: ("elevation" OR "altitude") AND ("anemia" OR "anaemia") AND ("prevalence" OR "DALYs" OR "YLDs"), examining all studies published between Jan 1, 2024, (the year of publication of the updated WHO guidelines) and Jan 13, 2026. We identified several assessments of the implications of the new WHO method, mostly limited to specific age ranges (eg, children aged 6 months to 5 years) or specific countries. One analysis by Luo and colleagues (2025) found that anaemia prevalence for children aged 6–59 months increased 4.7 percentage points (ranging from –0.4 to 18.1) using updated haemoglobin cutoff thresholds and the new elevation adjustment method, and anaemia prevalence for non-pregnant females aged 15–49 years increased from 13.5% to 21.3%. We identified no study assessing the global effect of changes to elevation adjustment across all demographics.

Added value of this study

To our knowledge, this study represents the first comprehensive comparison of the WHO 2001 and WHO 2024 haemoglobin elevation adjustment methods, assessing their effect on both anaemia prevalence and years lived with disability (YLDs) globally. Using data from GBD 2023, we evaluated levels and trends in anaemia prevalence and YLDs using both the WHO 2001 and WHO 2024 elevation adjustment methods for 204 countries and territories, 25 age groups, and male and female sexes from 1990 to 2023. Our results provide new insights into how updated elevation adjustment methods reshape global and regional understanding of anaemia burden, and highlight important age, sex, and geographical variations in effect.

Implications of all the available evidence

The WHO 2024 elevation adjustment method results in higher estimates of anaemia prevalence globally, emphasising the need for updated adjustment strategies based on diverse geographical contexts. Accurate estimates of anaemia prevalence are crucial for monitoring progress towards international reduction targets, as highlighted by extensive research efforts such as BRINDA. Assessment of the effect of different elevation adjustment methods is crucial for improving the accuracy of anaemia diagnosis and treatment worldwide. With improved methods for elevation adjustments now in place, future efforts can more effectively focus on evaluating the utility of adjustment recommendations specific to regional and demographic differences, enhancing methodologies for data collection and analysis, and revisiting assessments of anaemia prevention and treatment, especially in areas where anaemia has been historically underestimated or overestimated.

In 2001, WHO published anaemia definitions that included adjustment of haemoglobin for elevation.⁴ This adjustment equation was derived from the US Centers for Disease Control and Prevention in 1989 using data from a small, demographically limited subset of the global population.^{9–11} However, recent findings from the Biomarkers Reflecting Inflammation and Nutritional Determinants of Anemia (BRINDA) project indicate that the 2001 adjustment method probably under-adjusts haemoglobin concentrations at lower elevations and over-adjusts them at higher elevations.¹² Using a recent, more diverse dataset, BRINDA proposed a new adjustment equation, now adopted in the 2024 WHO anaemia assessment guidelines.³

In this study, we simultaneously generated updated estimates of anaemia burden and quantified the effect of changes between 2001 and 2024 WHO definitions, including changes to elevation adjustment and age-specific thresholds for young children. Understanding the implications of these adjustments will better inform

research and policy guidelines for producing high-quality and consistent estimates of anaemia burden.

Methods

Overview

We generated parallel estimates according to WHO 2001 and WHO 2024 elevation adjustment methods using the GBD approach, as previously published and detailed in the appendix (pp 5–18).¹ GBD 2021 anaemia sources¹ were supplemented with new years of data from the WHO Vitamin and Mineral Nutrition Information System up to December, 2022. We completed parallel data processing and estimation of two unique, continuous distributions of haemoglobin concentrations (g/L), anaemia prevalence, and unadjusted years lived with disability (YLDs) from 1990 to 2023 for 204 countries and territories, males and females, and 25 age groups (0–6 days, 7–27 days, 1–5 months, 6–11 months, 12–23 months, 2–4 years, 5–94 years in 5-year age bins, and ≥95 years). This study complies with GATHER¹³ (appendix pp 3–4). Ethics approval was not required, as

all analyses used publicly available, aggregated data. Analyses were performed in R (version 4.4.2) and Python (version 3.10).

Data extraction and processing

Input sources were included only if they either had quantitative, individual-level haemoglobin measurements (ie, microdata) or reported a measure of anaemia prevalence with a specified adjustment method in a representative population. All input sources are listed in the appendix (pp 23–99) and at the Global Health Data Exchange. The source-location-years available for both individual-level data and tabulated data are shown in the appendix (p 7).

All individuals with microdata were first assigned to a specific location, age, sex, and pregnancy status according to the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) 2023 demographics hierarchy. Survey-provided elevations were used if available. If not available, we used the most granular GBD location to assign each individual a population-weighted mean elevation (figure 1A) derived from combining 5×5 km resolution elevation¹⁴ and population raster files from 1980 to 2021.¹⁵ Elevation adjustments from each of the WHO 2001 and WHO 2024 definitions (figure 1B, C) were subtracted from raw haemoglobin values according to elevation at each person's reported residence. No additional adjustments were made for smoking status, method of haemoglobin sampling (eg, whole blood *vs* capillary), or analysis method (eg, Coulter counter *vs* point-of-care testing). Severity levels (mild, moderate, and severe) were then assigned using specific thresholds that vary by age, sex, and pregnancy status (figure 1D).

Studies containing only tabulated data were first flagged for the elevation adjustment method used to report the mean haemoglobin concentration and anaemia prevalence. Within each tabulated input data source, if any of the mean haemoglobin concentrations or total, moderate plus severe, or severe anaemia prevalence values were missing for a given age, sex, location, or year combination, those metrics were imputed using meta-regression—Bayesian, regularised, trimmed models (MR-BRT).¹⁶ The model used either the reported mean haemoglobin to estimate the missing total, moderate plus severe, or severe anaemia prevalence; or the reported total or moderate plus severe anaemia prevalence to estimate the missing mean haemoglobin concentration. This was done to ensure internal consistency across both the WHO 2001 and WHO 2024 elevation-adjusted datasets, such that every study had a datapoint for all measures estimated in our models. More detailed processing information can be found in the appendix (pp 8–15, 100–1168). The final output of this process was two parallel datasets—one each for the WHO 2001 and WHO 2024 elevation adjustment methods—containing adjusted mean haemoglobin and severity-specific anaemia prevalence observations from each identified data source.

Estimation of mean haemoglobin and severity-specific anaemia prevalence

We estimated log-transformed mean haemoglobin concentration and logit-transformed prevalence of severe, moderate plus severe, and total anaemia using spatiotemporal Gaussian process regression (ST-GPR) models.¹⁷ We first estimated a linear prior using covariates known to be biologically or epidemiologically associated with anaemia (appendix p 16). Next, we calculated the residuals between our model and our input data and smoothed these residuals over space, age, and time, producing a revised estimate for every location, year, age, and sex. The final step was a Gaussian process regression that further smoothed the residuals between our data and step two estimates, from which we quantified uncertainty in our final model estimates by taking 1000 draws from the posterior Gaussian process.

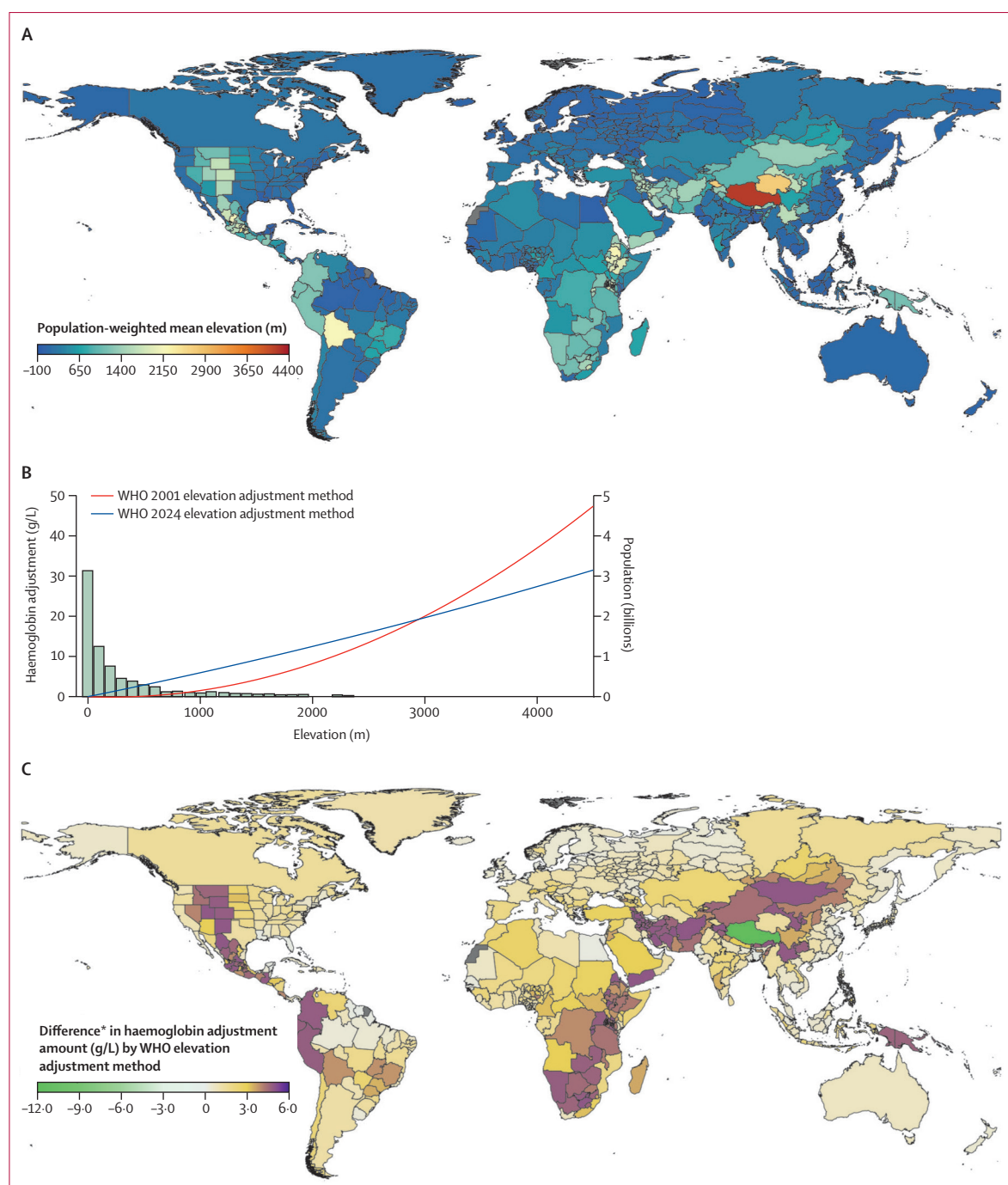
Generation of a continuous distribution of haemoglobin concentration

To derive the form of the final haemoglobin distributions, rather than assume a specific distribution shape *a priori*, we estimated an ensemble distribution consisting of a combination of two-parameter distribution families for each modelled age, sex, location, and year. The ensemble distribution was generated by using both single candidate two-parameter distributions (gamma, mirrored gamma, Weibull, mirrored lognormal, and mirrored Gumbel) and weighted ensembles of those same candidate distributions that best fit individual-level haemoglobin concentrations in our input data using out-of-sample fit metrics (appendix pp 16–17).

Calculation of anaemia prevalence and years lived with disability

By combining the mean haemoglobin concentrations obtained from the ST-GPR modelling phase and the final ensemble distribution weights, we calculated a variance for each combination of age, sex, year, and location. This variance was chosen to minimise the differences between the ST-GPR-estimated total, moderate plus severe, and severe anaemia prevalence estimates and the corresponding severity-specific anaemia prevalences derived from the probability density curve of the haemoglobin ensemble distribution. Next, using the ST-GPR estimated mean haemoglobin value, the ensemble distribution weights, and the optimised variance, we re-generated the haemoglobin distribution and calculated the area under the probability density curve between the haemoglobin thresholds to derive mild, moderate, and severe anaemia prevalence. Anaemia prevalence estimates by pregnancy status were aggregated to create estimates for all females aged 10–54 years. We used unpaired *t* tests to assess statistically significant differences between elevation adjustment methods to determine those results that are statistically

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(Figure 1 continues on next page)

meaningfully different when accounting for the overall uncertainty in the estimates and the difference attributable to changes in the elevation adjustment method.

Unadjusted YLDs were calculated by multiplying the prevalence of mild, moderate, and severe anaemia by their respective GBD disability weights (where a disability weight of 0 is no health loss and a disability weight of 1 is death).

The disability weights were 0.004 for mild anaemia, 0.052 for moderate anaemia, and 0.149 for severe anaemia.¹ The YLD metric allows for standardised comparisons of non-fatal disease burden between severities of anaemia.

Paediatric threshold analysis

The WHO 2024 anaemia assessment guidelines included updated haemoglobin thresholds for children aged

6–23 months.³ To assess the combined effect of both the elevation adjustment and threshold changes, we performed a secondary analysis by re-calculating anaemia prevalence for all individual-level data sources using three different combinations: WHO 2001 elevation adjustment with WHO 2001 thresholds, WHO 2024 elevation adjustment with WHO 2001 thresholds, and WHO 2024 elevation adjustment with WHO 2024 thresholds for children aged 6–23 months. We calculated average differences in anaemia prevalence across all location-years to isolate the effects of threshold changes from elevation adjustment changes (appendix pp 20–21).

Epidemiological transition analysis

To evaluate whether the elevation adjustment method led to changes in the historical relationship between anaemia burden and sociodemographic development, we performed an epidemiological transition analysis comparing the results for the two adjustment methods. Briefly, we performed an ecological analysis between Socio-demographic Index (SDI)¹⁸ and age-specific and sex-specific total, moderate plus severe, and severe anaemia prevalence across all GBD locations and years using MR-BRT models.

The input data for the model were created by matching post-ensemble age-specific and sex-specific anaemia prevalences with location-year-specific SDI estimates. Prevalence data were logit transformed, and all locations were given an equal weight in the meta-regression (appendix pp 18–19).

Uncertainty

Uncertainty was propagated in each step of our modelling process, including sample variance from the input data and uncertainty generated in each of the MR-BRT, ST-GPR, and post-ensemble models. We estimated uncertainty by sampling draws from the posterior distribution of each estimated quantity. Reported upper and lower uncertainty intervals (UIs) correspond to the 2·5th and 97·5th percentile of the draws, respectively. Aggregations by geography, age, and sex were made at the draw level, assuming uncorrelated uncertainty.

Role of the funding source

The funder had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Global anaemia prevalence in 2023 aggregated across all ages and both sexes was 24·0% (95% UI 20·6–29·3) for the WHO 2001 elevation adjustment method and 26·4% (22·6–31·9) for the WHO 2024 elevation adjustment method (a difference of 2·4 percentage points [2·2–2·7]), making anaemia the second largest cause of disability in 2023. This corresponds to an estimated 198 million (196–199) greater number of anaemia cases globally using

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Sex	Age	Blood haemoglobin concentration (g/L)		
		Mild anaemia	Moderate anaemia	Severe anaemia
Both	0–6 days	145–159	100–144	<100
Both	7–27 days	120–134	85–119	<85
Both	1–5 months	100–109	70–79	<70
Both	6–23 months	95–104	70–94	<70
Both	2–4 years	100–109	70–99	<70
Both	5–9 years	110–114	80–109	<80
Both	10–14 years	110–119	80–109	<80
Male	≥15 years	110–129	80–109	<80
Female, non-pregnant	≥15 years	110–119	80–109	<80
Female, pregnant	≥10 years	100–109	70–99	<70

Figure 1: Definitions of anaemia cutoff concentration values and population-weighted mean elevations (A) Population-weighted mean elevations by modelled locations. (B) Haemoglobin adjustment for elevation. Population distribution is added to illustrate the number of people most affected by the differences in elevation adjustment methods. (C) Difference in haemoglobin adjustment between the WHO 2001 and WHO 2024 elevation adjustment methods. (D) Definitions of mild, moderate, and severe anaemia based on blood haemoglobin concentration. *The values represent the WHO 2001 adjustment amount subtracted from the WHO 2024 adjustment amount.

the WHO 2024 method than with the WHO 2001 method (table). The WHO 2024 method resulted in an increase in moderate plus severe anaemia prevalence of 1·5 percentage points (1·4–1·5) and an increase in severe anaemia prevalence of 0·2 percentage points (0·2–0·3). This equates to an estimated 120 million (119–121) more cases of moderate plus severe anaemia and 19·2 million (19·1–19·3) more cases of severe anaemia globally in all ages and both sexes in 2023 when using the WHO 2024 method. Comparisons of all annual age-sex-specific trends from 1990 to 2023 are in the appendix (pp 1169–1990). Selected years from 1990 to 2023 of severity-specific prevalence and YLDs for all ages and males and females combined are in the appendix (pp 1993–2413).

Total anaemia prevalence using both adjustment methods and the absolute difference between adjustment methods is shown in figure 2. There were disparities in total anaemia prevalence by geographical region, irrespective of the elevation adjustment method used. Total anaemia prevalence for all ages and both sexes in 2023, based on the WHO 2024 elevation adjustment method, ranged from 7·6% (95% UI 5·6–10·3) in Scotland to 58·7% (53·2–64·6) in Yemen (figure 2B). Overall, the WHO 2024 method resulted in a higher anaemia prevalence across all severities for all ages, sexes, and locations with a population-weighted mean elevation below 3000 m (figure 2C, D). The countries with the highest total anaemia prevalence for all ages and both sexes in 2023 were predominantly in sub-Saharan Africa (figure 2A, B). For example, Mali had a total anaemia prevalence of 57·1% (44·6–71·6) using the WHO 2024 method. Subnational units within Nigeria had the highest overall total anaemia prevalence when using the WHO 2024 method—eg, the state of Jigawa, Nigeria, had

	Mean haemoglobin (g/L)	Prevalence per 100 000 population			Total YLDs per 100 000 population	Prevalence, thousands			Total YLDs
		Total anaemia	Moderate plus severe anaemia	Severe anaemia		Total anaemia	Moderate plus severe anaemia	Severe anaemia	
Global									
WHO 2024 elevation adjustment method	134 (130–138)	26 400 (22 600–31 900)	11 800 (9 440–14 500)	1260 (740–2110)	794 (630–1010)	2 130 000 (1 820 000–2 580 000)	951 000 (762 000–1 170 000)	101 000 (59 700–170 000)	64 000 (50 800–81 100)
WHO 2001 elevation adjustment method	136 (132–140)	24 000 (20 600–29 300)	10 300 (8 210–12 900)	1020 (592–1860)	689 (556–884)	1 930 000 (1 660 000–2 370 000)	831 000 (662 000–1 040 000)	82 300 (47 700–150 000)	55 600 (44 800–71 300)
Central Asia									
WHO 2024 elevation adjustment method	133 (126–139)	28 200 (21 800–38 700)	12 000 (8 490–16 300)	1090 (351–2420)	793 (556–1140)	28 200 (21 700–38 600)	11 900 (8480–16 300)	1090 (351–2420)	792 (555–1130)
WHO 2001 elevation adjustment method	135 (129–142)	24 000 (18 600–32 700)	10 100 (7010–14 300)	920 (302–2150)	669 (463–1010)	24 000 (18 600–32 600)	10 100 (7010–14 300)	919 (302–2150)	668 (462–1010)
Central Europe									
WHO 2024 elevation adjustment method	141 (132–150)	15 400 (9870–23 200)	4630 (2000–8840)	357 (45–6–1160)	319 (147–608)	17 900 (11 500–26 900)	5380 (2330–10 300)	414 (52–9–1350)	370 (170–706)
WHO 2001 elevation adjustment method	141 (133–150)	13 400 (8550–20 600)	3880 (1640–7480)	287 (40–3–897)	268 (128–474)	15 600 (9920–23 900)	4500 (1910–8680)	333 (46–8–1040)	311 (149–550)
Eastern Europe									
WHO 2024 elevation adjustment method	142 (135–150)	15 300 (10 700–21 600)	5700 (3310–9460)	552 (127–1490)	389 (219–680)	31 700 (22 200–44 800)	11 800 (6860–19 600)	1140 (264–3090)	807 (453–1410)
WHO 2001 elevation adjustment method	143 (135–150)	13 900 (9780–20 100)	5020 (2850–8360)	465 (111–1280)	343 (202–563)	28 900 (20 300–41 600)	10 400 (5900–17 300)	963 (229–2660)	710 (418–1170)
Australasia									
WHO 2024 elevation adjustment method	143 (139–147)	10 600 (6760–16 600)	3110 (1480–6330)	247 (62–7–722)	216 (108–440)	3390 (2170–5340)	998 (475–2030)	79·3 (20·1–232)	69·2 (34·7–141)
WHO 2001 elevation adjustment method	143 (139–148)	9670 (6180–15 400)	2760 (1360–5580)	218 (58–6–624)	192 (98–2–376)	3100 (1980–4920)	886 (436–1790)	70·1 (18·8–200)	61·7 (31·5–121)
High-income Asia Pacific									
WHO 2024 elevation adjustment method	143 (139–146)	14 300 (12 200–17 200)	4660 (3550–6280)	418 (204–797)	321 (248–446)	26 200 (22 300–31 500)	8500 (6490–11 500)	763 (372–1460)	587 (452–815)
WHO 2001 elevation adjustment method	143 (140–146)	13 500 (11 200–16 300)	4120 (2920–5770)	344 (152–696)	285 (205–394)	24 600 (20 400–29 700)	7510 (5320–10 500)	627 (277–1270)	520 (375–720)
High-income North America									
WHO 2024 elevation adjustment method	140 (138–142)	11 300 (9800–13 200)	3030 (2480–3810)	186 (106–319)	209 (173–264)	42 200 (36 600–49 500)	11 300 (9260–14 200)	697 (397–1190)	781 (648–989)
WHO 2001 elevation adjustment method	141 (140–143)	10 000 (8630–11 600)	2700 (2220–3310)	171 (107–269)	186 (158–231)	37 500 (32 300–43 500)	10 100 (8290–12 400)	641 (399–1000)	697 (590–862)
Southern Latin America									
WHO 2024 elevation adjustment method	140 (134–146)	12 500 (7630–18 900)	3560 (1380–7210)	292 (51–6–882)	249 (103–506)	8770 (5340–13 200)	2490 (967–5050)	205 (36–1–618)	174 (72–2–354)
WHO 2001 elevation adjustment method	141 (135–148)	11 300 (6850–17 400)	3050 (1150–6200)	244 (38–1–751)	215 (91–0–403)	7940 (4800–12 200)	2130 (807–4340)	171 (26–7–526)	151 (63–8–282)
Western Europe									
WHO 2024 elevation adjustment method	144 (140–149)	9930 (7450–13 500)	2920 (1910–4860)	254 (111–564)	204 (135–336)	44 200 (33 200–60 300)	13 000 (8480–21 600)	1130 (493–2510)	910 (599–1490)
WHO 2001 elevation adjustment method	144 (140–149)	9020 (6690–12 400)	2530 (1600–4180)	209 (91–8–488)	178 (115–277)	40 200 (29 800–55 100)	11 200 (7100–18 600)	928 (409–2170)	790 (512–1230)
Andean Latin America									
WHO 2024 elevation adjustment method	135 (131–140)	20 100 (15 900–26 100)	7230 (5170–10 000)	563 (242–1140)	482 (335–674)	13 300 (10 500–17 200)	4770 (3410–6 630)	372 (159–753)	318 (221–445)
WHO 2001 elevation adjustment method	139 (135–142)	16 600 (13 100–20 800)	6190 (4370–8920)	515 (233–1110)	413 (298–601)	10 900 (8650–13 700)	4080 (2880–5890)	340 (154–730)	272 (196–396)
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	Mean haemoglobin (g/L)	Prevalence per 100 000 population			Total YLDs per 100 000 population	Prevalence, thousands			Total YLDs
		Total anaemia	Moderate plus severe anaemia	Severe anaemia		Total anaemia	Moderate plus severe anaemia	Severe anaemia	
(Continued from previous page)									
Caribbean									
WHO 2024 elevation adjustment method	132 (126–139)	28 800 (22 200–37 700)	12 400 (7930–18 200)	1270 (480–2980)	832 (516–1310)	13 900 (10 800–18 300)	5990 (3840–8840)	617 (232–1440)	403 (250–634)
WHO 2001 elevation adjustment method	133 (127–139)	27 300 (21 200–36 700)	11 200 (7260–16 600)	1060 (407–2570)	751 (510–1140)	13 200 (10 300–17 800)	5440 (3520–8030)	512 (197–1250)	364 (247–552)
Central Latin America									
WHO 2024 elevation adjustment method	140 (135–145)	17 300 (13 500–22 200)	6830 (4810–9810)	695 (285–1520)	465 (320–693)	44 900 (35 100–57 600)	17 700 (12 500–25 400)	1800 (739–3940)	1210 (830–1800)
WHO 2001 elevation adjustment method	143 (138–148)	13 700 (10 800–17 600)	5180 (3650–7570)	524 (238–1 080)	354 (245–506)	35 500 (27 900–45 600)	13 400 (9 470–19 600)	1360 (616–2 800)	918 (635–1 310)
Tropical Latin America									
WHO 2024 elevation adjustment method	137 (133–141)	16 200 (13 400–21 900)	5760 (4470–7190)	528 (253–1010)	392 (293–512)	35 300 (29 400–47 800)	12 600 (9770–15 700)	1150 (553–2200)	857 (639–1120)
WHO 2001 elevation adjustment method	139 (135–143)	13 300 (10 900–17 000)	4620 (3440–5890)	402 (188–738)	314 (234–400)	29 000 (23 900–37 100)	10 100 (7500–12 900)	879 (410–1610)	685 (510–875)
North Africa and Middle East									
WHO 2024 elevation adjustment method	135 (130–139)	24 700 (21 100–30 000)	10 600 (8760–13 300)	1260 (894–2070)	731 (600–966)	158 000 (135 000–192 000)	67 700 (55 900–84 700)	8070 (5710–13 200)	4660 (3830–6160)
WHO 2001 elevation adjustment method	136 (132–141)	21 500 (18 200–26 500)	8940 (7240–11 500)	1000 (698–1720)	612 (500–766)	137 000 (116 000–169 000)	57 000 (46 200–73 400)	6380 (4450–11 000)	3910 (3190–4890)
South Asia									
WHO 2024 elevation adjustment method	126 (121–131)	41 800 (33 600–52 900)	20 400 (15 400–26 000)	2270 (1060–4030)	1370 (1030–1780)	791 000 (637 000–1 000 000)	387 000 (291 000–493 000)	43 000 (20 100–76 300)	25 900 (19 500–33 700)
WHO 2001 elevation adjustment method	128 (123–132)	39 400 (32 200–50 000)	18 500 (13 900–23 500)	1860 (855–3570)	1230 (922–1630)	746 000 (610 000–946 000)	350 000 (263 000–446 000)	35 300 (16 200–67 600)	23 200 (17 500–30 800)
East Asia									
WHO 2024 elevation adjustment method	144 (137–151)	11 600 (9570–14 300)	3700 (2730–4740)	305 (112–564)	253 (188–321)	172 000 (142 000–212 000)	54 700 (40 500–70 200)	4520 (1650–8360)	3750 (2790–4760)
WHO 2001 elevation adjustment method	146 (139–153)	10 600 (8850–13 000)	3390 (2610–4310)	298 (127–537)	234 (183–304)	156 000 (131 000–193 000)	50 100 (38 600–63 800)	4410 (1880–7940)	3460 (2700–4490)
Oceania									
WHO 2024 elevation adjustment method	129 (126–132)	33 200 (28 500–40 700)	14 500 (12 000–17 200)	1510 (743–2520)	979 (778–1210)	4900 (4210–6010)	2140 (1770–2540)	223 (110–372)	145 (115–178)
WHO 2001 elevation adjustment method	132 (129–135)	27 700 (23 800–32 800)	11 900 (9090–14 500)	1210 (573–2130)	799 (593–1010)	4080 (3510–4840)	1750 (1340–2140)	178 (84.6–314)	118 (87.5–148)
Southeast Asia									
WHO 2024 elevation adjustment method	135 (132–137)	24 400 (21 800–28 600)	9580 (8140–11 300)	848 (519–1340)	640 (536–757)	174 000 (156 000–204 000)	68 400 (58 100–80 400)	6050 (3710–9560)	4570 (3830–5410)
WHO 2001 elevation adjustment method	136 (133–138)	22 600 (20 000–26 100)	8430 (6890–10 300)	694 (420–1160)	562 (451–696)	161 000 (143 000–187 000)	60 200 (49 200–73 600)	4950 (3000–8290)	4010 (3220–4970)
Central sub-Saharan Africa									
WHO 2024 elevation adjustment method	124 (121–127)	42 400 (36 000–51 500)	20 800 (17 000–24 600)	2260 (1270–3660)	1380 (1140–1680)	62 200 (52 800–75 400)	30 400 (24 900–36 100)	3310 (1850–5370)	2030 (1680–2460)
WHO 2001 elevation adjustment method	127 (124–130)	35 800 (30 700–44 900)	16 700 (13 600–20 300)	1630 (842–2910)	1100 (915–1360)	52 500 (44 900–65 700)	24 400 (19 900–29 800)	2390 (1230–4260)	1620 (1340–1990)
Eastern sub-Saharan Africa									
WHO 2024 elevation adjustment method	125 (121–128)	40 300 (34 500–47 500)	20 500 (16 600–24 900)	2510 (1630–3970)	1390 (1120–1740)	182 000 (156 000–215 000)	92 900 (75 200–113 000)	11 400 (7360–17 900)	6290 (5050–7870)
WHO 2001 elevation adjustment method	128 (125–132)	33 700 (28 900–40 500)	16 400 (13 200–20 000)	1840 (1120–3130)	1100 (896–1400)	152 000 (131 000–183 000)	74 000 (59 800–90 300)	8310 (5060–14 200)	4970 (4050–6330)
(Table continues on next page)									

(Table continues on next page)

	Mean haemoglobin (g/L)	Prevalence per 100 000 population			Total YLDs per 100 000 population	Prevalence, thousands			Total YLDs
		Total anaemia	Moderate plus severe anaemia	Severe anaemia		Total anaemia	Moderate plus severe anaemia	Severe anaemia	
(Continued from previous page)									
Southern sub-Saharan Africa									
WHO 2024 elevation adjustment method	130 (127–133)	33 100 (28 600–39 800)	14 200 (10 500–18 000)	1380 (648–2530)	948 (701–1270)	29 300 (25 300–35 200)	12 500 (9340–15 900)	1220 (574–2240)	840 (621–1120)
WHO 2001 elevation adjustment method	133 (131–136)	25 900 (22 000–30 700)	10 800 (8070–14 200)	1040 (520–1960)	725 (534–981)	22 900 (19 500–27 200)	9570 (7150–12 600)	920 (460–1740)	642 (473–869)
Western sub-Saharan Africa									
WHO 2024 elevation adjustment method	121 (117–126)	47 800 (38 800–60 000)	24 700 (18 600–31 100)	2780 (1420–4880)	1650 (1240–2090)	248 000 (202 000–312 000)	129 000 (96 600–162 000)	14 500 (7380–25 300)	8570 (6450–10 900)
WHO 2001 elevation adjustment method	123 (118–127)	44 300 (36 100–56 800)	21 900 (16 200–28 400)	2250 (1140–4400)	1450 (1070–1880)	230 000 (188 000–295 000)	114 000 (84 200–147 000)	11 700 (5940–22 900)	7520 (5580–9760)
Data in parentheses are 95% uncertainty intervals. Count data are presented to three significant figures. YLDs=years lived with disability.									
Table: Total anaemia and mild, moderate, and severe anaemia prevalence and cases using WHO 2001 versus WHO 2024 elevation adjustment methods, 2023									

Data in parentheses are 95% uncertainty intervals. Count data are presented to three significant figures. YLDs=years lived with disability.

Table: Total anaemia and mild, moderate, and severe anaemia prevalence and cases using WHO 2001 versus WHO 2024 elevation adjustment methods, 2023

an estimated total anaemia prevalence of 67.3% (49.2–80.2). Western European high-income countries and their respective subnational units were among the locations with the lowest total anaemia prevalence in 2023 for the same demographics. Within the western European high-income countries, Greece had the highest total anaemia prevalence of 13.5% (10.1–18.4).

Countries with population-weighted mean elevations between 500 m and 2000 m had the largest differences in total anaemia prevalence by WHO elevation adjustment method for all ages and both sexes in 2023 (figure 2C, D). This included mountainous areas in eastern sub-Saharan Africa, central Asia, and countries in central America and South America. Burundi had the largest difference in total anaemia prevalence between the elevation adjustment methods, where the WHO 2024 method measured a total anaemia prevalence for all ages and both sexes that was 9.6 percentage points (95% UI 9.4–9.7) higher than the total anaemia prevalence using the WHO 2001 method. Tibet (province of China), with a population-weighted mean elevation above 3000 m, had an estimated total anaemia prevalence of 4.7 percentage points (4.7–4.7) lower using the WHO 2024 method than the WHO 2001 method.

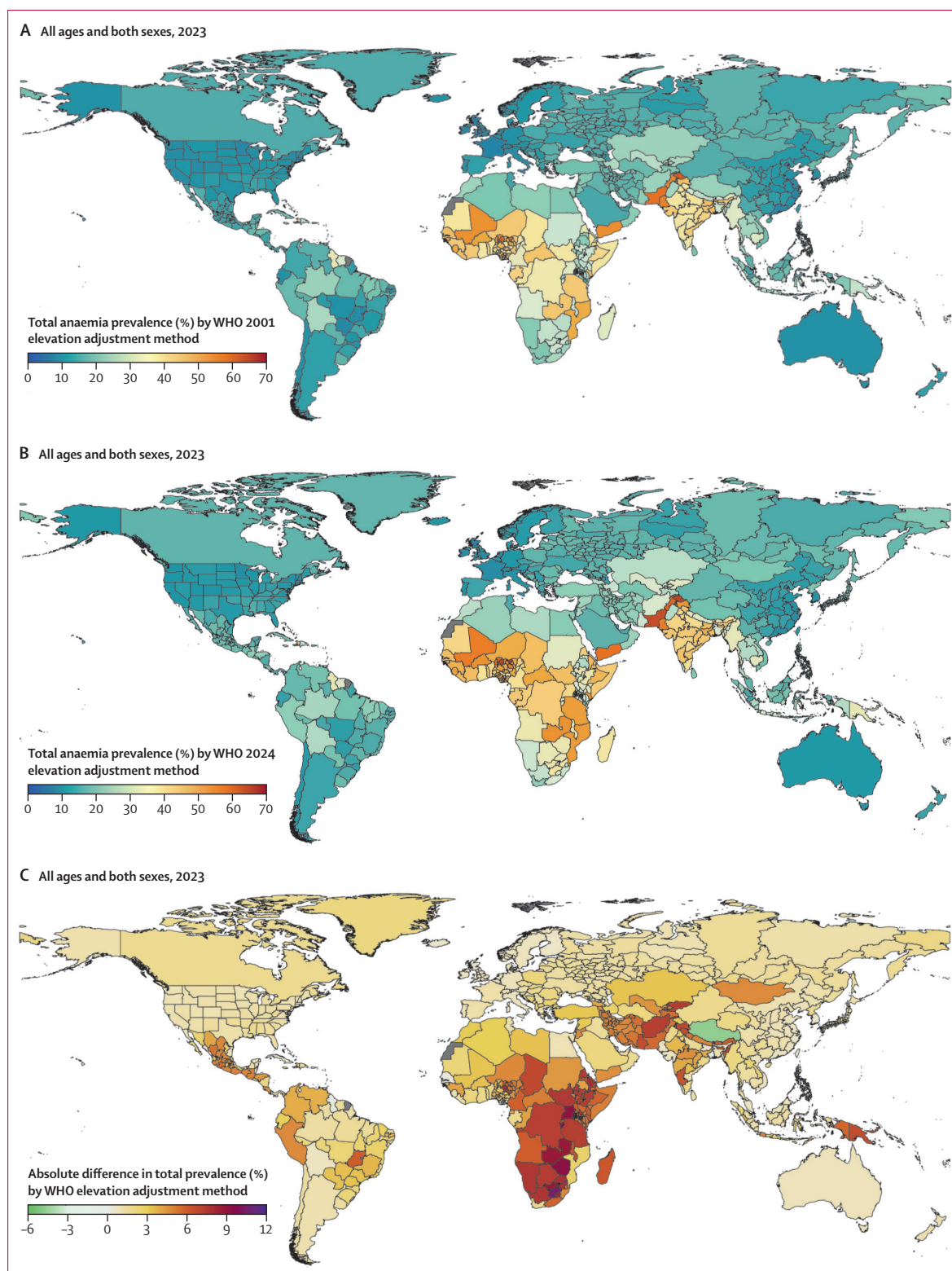
In 2023, female all-age total anaemia prevalence when using the WHO 2024 elevation adjustment method was 31.0% (95% UI 26.9–37.7), whereas male all-age total anaemia prevalence was 21.9% (18.2–26.7). This corresponds to an unadjusted YLD rate per 100 000 for total anaemia of 1020 (817–1250) for females and 566 (442–766) for males for all ages in 2023 globally, which ranks anaemia as second and fifth overall for causes of disability, respectively.

Among all demographics, females aged 10–54 years (ie, women of reproductive age) and children younger than 5 years had the largest burden of anaemia when using the

WHO 2024 elevation adjustment method, with a global total anaemia prevalence in 2023 of 31.9% (95% UI 28.2–38.3) in females aged 10–54 years (appendix pp 2134–2273) and 39.2% (35.0–45.4) in children younger than 5 years (appendix pp 2274–2413). When comparing global total anaemia prevalence estimates in 2023, we estimated increases of 3.0 percentage points (3.0–3.0) for females aged 10–54 years and 3.9 percentage points (3.9–3.9) for children younger than 5 years when using the WHO 2024 versus the WHO 2001 elevation adjustment method. This corresponded to 75.0 million (74.6–75.5) and 23.4 million (23.2–23.6) more cases globally in 2023 for each of these groups, respectively.

Geographical trends for differences in total anaemia prevalence between the two WHO elevation adjustment methods for females aged 10–54 years and children younger than 5 years followed that of the total anaemia prevalence for all ages and both sexes (figure 3). The locations with the largest differences in total anaemia prevalence when comparing the WHO 2024 and WHO 2001 elevation adjustment methods were Northern Cape, South Africa (2156 m), with an increase of 15.4 percentage points (95% UI 15.2–15.5) for children younger than 5 years, and Gauteng, South Africa (1913 m), with an increase of 12.5 percentage points (12.4–12.5) for females aged 10–54 years.

Moderate plus severe and severe anaemia prevalence estimates for 2023 also showed larger increases in all demographics when using the WHO 2024 elevation adjustment method than with the WHO 2001 method for locations with a population-weighted mean elevation between 500 m and 2000 m (appendix pp 1993–2413). For example, in Somali (region of Ethiopia), we observed increases of up to 2.0 percentage points (95% UI 2.0–2.0) in our estimates of severe anaemia prevalence for females aged 10–54 years, which correlates to an



(Figure 2 continues on next page)

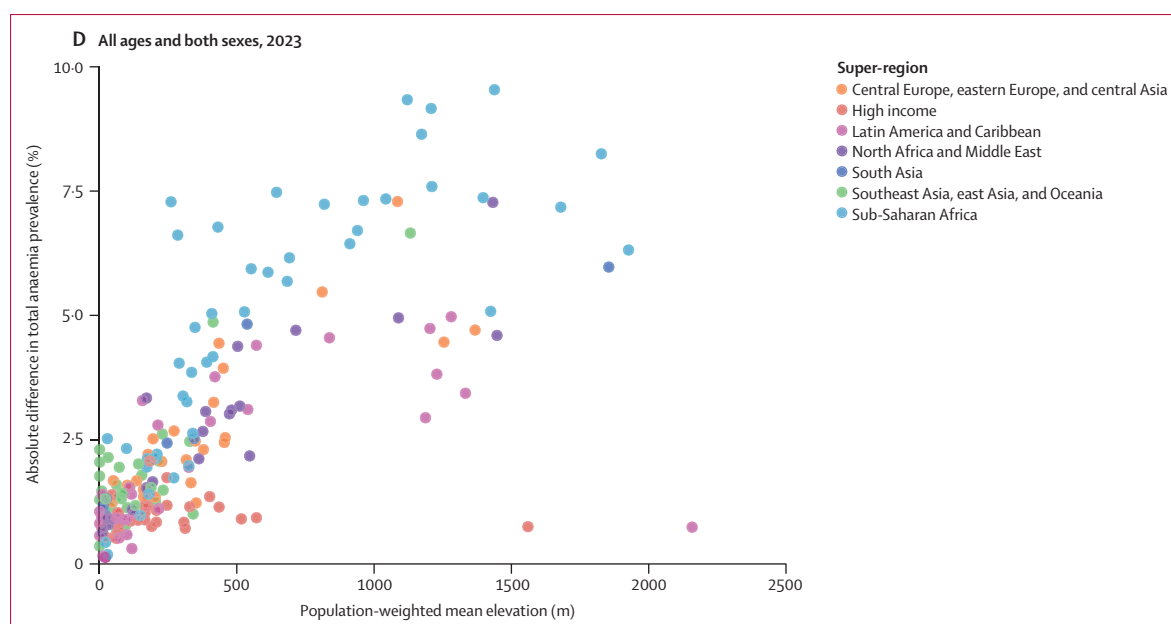


Figure 2: Comparison of total anaemia prevalence between the WHO 2001 and WHO 2024 elevation adjustment methods, 2023

(A) Total anaemia prevalence according to the WHO 2001 elevation adjustment method. (B) Total anaemia prevalence according to the WHO 2024 elevation adjustment method. (C) Absolute difference in total anaemia prevalence between the WHO 2024 and WHO 2001 elevation adjustment methods. (D) Absolute difference in total anaemia prevalence between the WHO 2024 and WHO 2001 elevation adjustment methods plotted against country population-weighted mean elevation.

increase of 294 (292–296) in unadjusted YLD rate per 100 000 population. In Yemen, estimates of severe anaemia prevalence for children younger than 5 years showed increases up to 4.4 percentage points (4.4–4.4), resulting in an increase of 651 (649–654) in unadjusted YLD rate per 100 000 population in 2023.

In our supplemental analysis of threshold changes in children aged 6–23 months, the WHO 2024 elevation adjustment method resulted in an average increase of 5.8 percentage points (95% UI 0.8–14.3) in total anaemia prevalence in this group across all data sources (appendix p 20). However, within the same WHO 2024 method, the updated paediatric haemoglobin thresholds led to an average decrease in anaemia prevalence of 11.6 percentage points (4.3–17.7). The combined effect of updated elevation adjustment and thresholds resulted in a 5.8 percentage point (0.6–13.3) decrease in anaemia prevalence. Thus, although the WHO 2024 method increased anaemia prevalence globally, revised thresholds resulted in a net reduction in the estimated number of anaemia cases in children aged 6–23 months (appendix p 21).

Accounting for elevation with the WHO 2024 adjustment method compared with the WHO 2001 method most significantly affected countries with low to middle SDI (figure 4A–C). We observed a negative correlation between SDI and elevation adjustment amount, meaning that countries with lower SDIs are those where we would expect to see the largest changes in haemoglobin adjustment amounts and, therefore, higher anaemia prevalence. Among countries with SDI

at or above the 75th percentile, both adjustment methods resulted in similar anaemia prevalences, regardless of severity. However, as SDI decreases, the difference in anaemia prevalence estimated by the WHO 2001 versus the WHO 2024 adjustment method increases, with the WHO 2024 method resulting in a significantly greater anaemia prevalence, particularly in countries with an SDI between the 25th and 75th percentiles. Results for select demographics comparing SDI and anaemia prevalence can be found in the appendix (pp 2414–2420).

Discussion

According to the WHO 2024 elevation adjustment method, more than 2.1 billion people had anaemia globally in 2023 compared with 1.9 billion according to the WHO 2001 method. This number was a nearly 200 million case increase and would have made anaemia rank second, compared with third, among GBD disability causes (after low back pain). Most newly enumerated cases were in children younger than 5 years and females aged 10–54 years, and were concentrated in countries in eastern sub-Saharan Africa, central Asia, and central and South America that are simultaneously at relatively high elevation, on the lower end of the SDI spectrum, and have comparatively high rates of many anaemia-causing diseases.¹ These changes translated to an even stronger gradient between SDI and anaemia, especially in lower-SDI settings, supporting a notion that sociodemographic factors of income, education, and women's empowerment are universally key anaemia drivers.

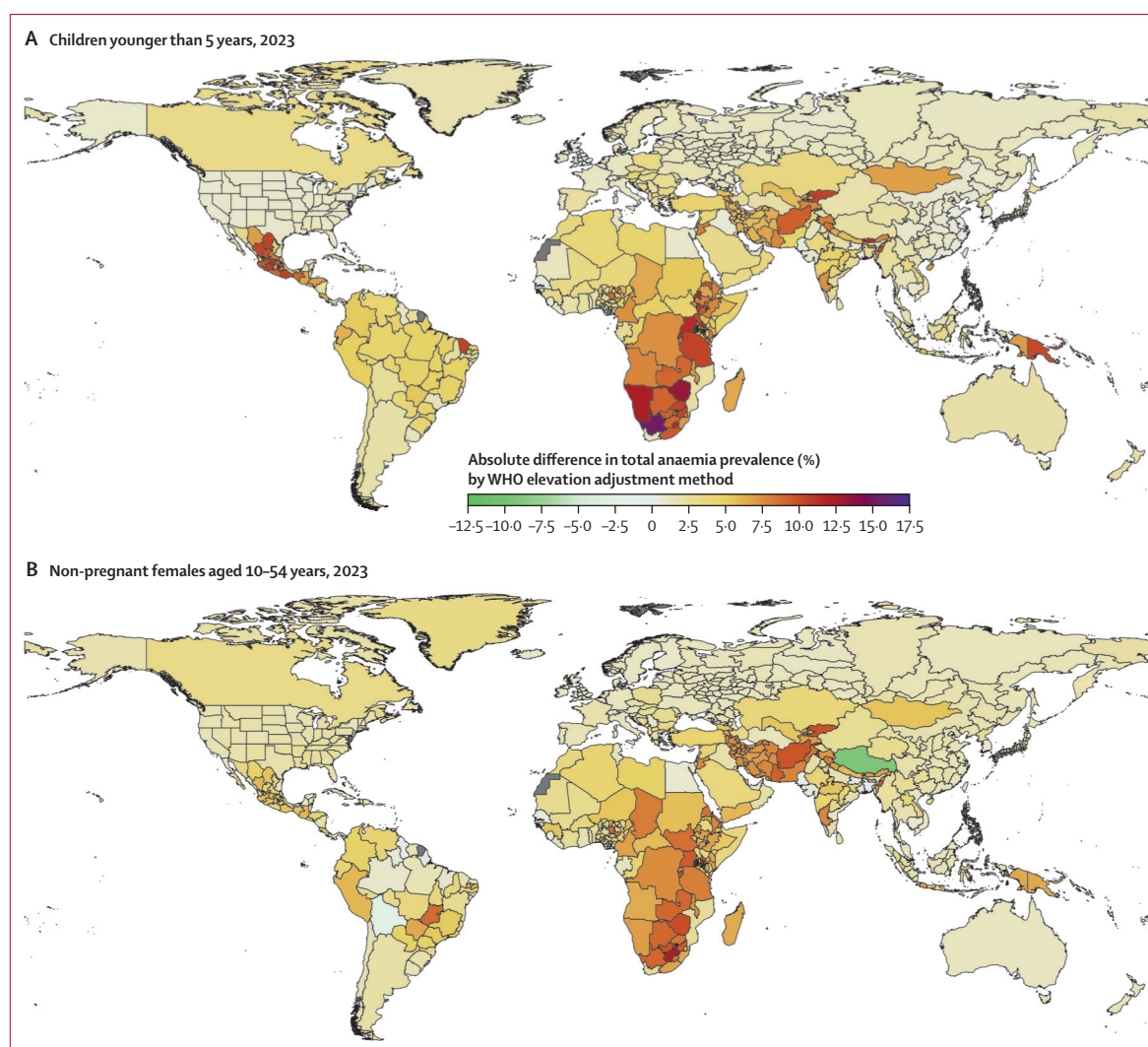


Figure 3: Absolute differences in anaemia prevalence between elevation adjustment methods in children younger than 5 years and non-pregnant females (aged 10–54 years), 2023

Absolute differences in total anaemia prevalence between the WHO 2024 and WHO 2001 elevation adjustment methods in 2023 for children younger than 5 years (A) and females of reproductive age (B).

There is little debate about the importance of adjusting haemoglobin for elevation,¹⁹ given that haemoglobin concentration is only a proxy for blood oxygen supply (the clinical definition of anaemia). Updates to WHO 2024 anaemia definitions facilitate improved monitoring of progress towards international targets and thereby better inform resource allocation for anaemia control.^{20–22} Examination of the process and assumptions used to derive both elevation adjustment methods underscores the appropriateness of the WHO 2024 update. The WHO 2001 elevation adjustment method was based on a comparatively small dataset with limited geographical and economic diversity (children and adult males from Chile, Peru, the UK, and the USA), consisting of data collected before the mid-1980s.^{9,10} According to the statistical methods used to calculate the WHO 2001

elevation adjustment equation,^{9,10} it appears that there was no attempt to account for demographics (eg, age and sex), and a log-linear functional form was assumed (figure 1B). By contrast, the WHO 2024 elevation adjustment, which leveraged the work of BRINDA,^{12,23} used a more comprehensive, multinational set of surveys, reflecting broader representation by age, sex, and location. BRINDA used biomarkers to identify a healthier subset of respondents (eg, removing participants with conditions that might be associated with iron deficiency and anaemia), completed several sensitivity analyses to assess heterogeneity in associations, and evaluated multiple statistical models before deriving a nearly linear quadratic functional form (figure 1B).^{12,23}

Although previous studies have reported on broader expected changes in the assessment of anaemia burden as

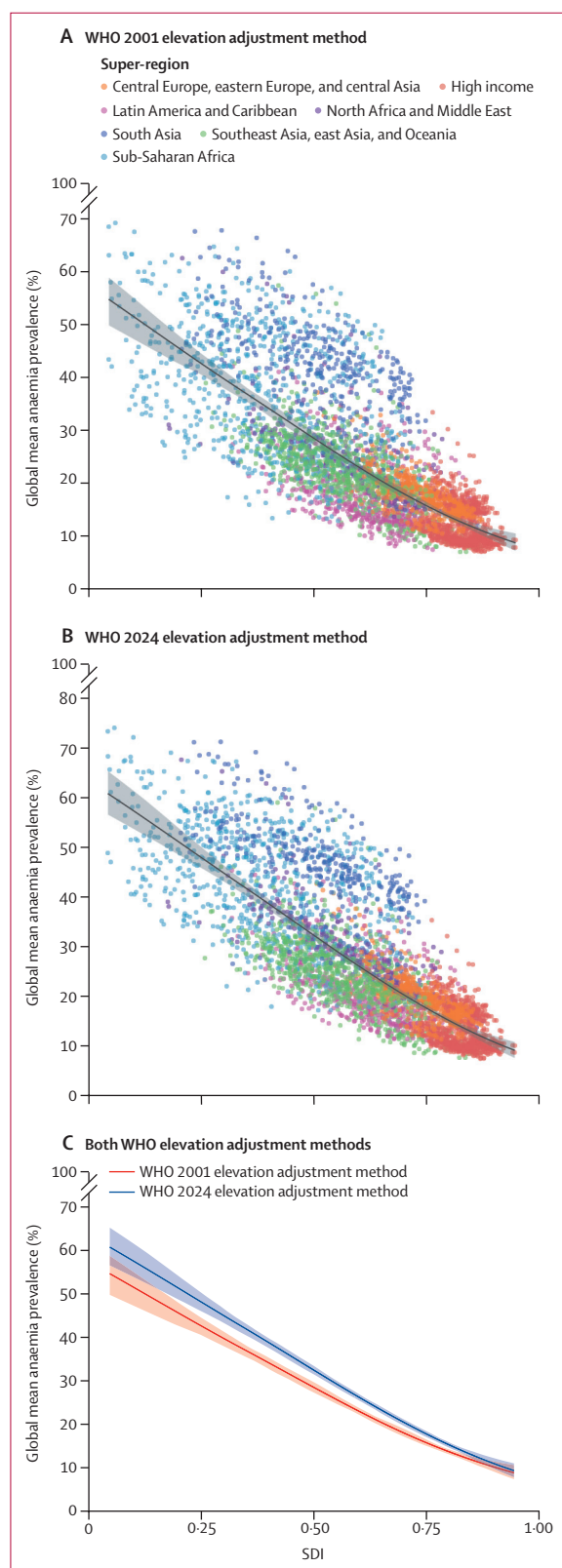


Figure 4: Modelled relationships of SDI and anaemia prevalence, 1990–2023

(A) Global mean prevalence of total anaemia by SDI according to the WHO 2001 elevation adjustment method. (B) Global mean prevalence of total anaemia by SDI according to the WHO 2024 elevation adjustment method. (C) Global mean prevalence of total anaemia by SDI according to the WHO elevation adjustment methods. Shaded areas indicate 95% uncertainty intervals. SDI=Socio-demographic Index.

a consequence of WHO 2024 definitional changes,^{19,24,25} our study is the first, to our knowledge, to comprehensively analyse the implications of definitional changes across locations, age, sex, and over time, including reprocessing of all 818 individual-level and tabulated datasets of haemoglobin concentration and anaemia prevalence. For children aged 6–23 months in particular, our secondary analysis of the effect of threshold changes found that elevation adjustment changes partially, but not completely, offset the decreases that accompanied threshold changes and that differences were most pronounced in mild and moderate anaemia (severe anaemia thresholds were unchanged and therefore unaffected). We were unable to assess the effect of WHO 2024 changes to trimester-specific thresholds for anaemia in pregnancy because our models currently evaluate pregnant people as a single group, but we expect that the population-level effect is likely to be modest because changes were only made for second-trimester thresholds.

Our study provides crucial insights, but it is not without limitations. First, as is true for all global analyses of anaemia burden, data availability varied by age and sex, with the greatest number of surveys of haemoglobin concentrations available in children younger than 5 years and females aged 10–54 years. More data need to be collected for adult males and individuals (both male and female) older than 60 years, along with potentially considering different adjustment approaches for populations residing at high elevations, given that these groups might have unique physiological adaptations to hypoxic environments.^{24,26} Second, we estimated anaemia cutoff concentrations for age groups younger than 6 months. Although these cutoff values are generated systematically, additional surveys focusing on infants in this age range are needed to enable WHO to establish more precise anaemia thresholds. Third, in cases for which elevation data were not provided in the original survey, our analysis used population-weighted mean elevations. This approach provides a general estimate, but it might not fully capture the variation in elevation within certain regions that we modelled. Although we assigned population-weighted mean elevations to the most granular locations available in the survey, further efforts could focus on isolating specific locations and elevations to improve accuracy. Fourth, we analysed haemoglobin concentrations collected using both venous and capillary methods, which are known to exhibit differing levels of variability in the amount of haemoglobin detected in the blood sample.²⁷ Relatedly, the method of analysis (eg, laboratory vs point-of-care testing) has also been shown to affect estimates of anaemia prevalence.²⁸ Because of scarce data availability to formally adjust for these differences, we pooled data using different blood sources and methods of analysis. Further investigation is needed to ensure that haemoglobin measurements from different methods can be reliably used and appropriately adjusted to account for these differences. Potential approaches

include developing a crosswalk adjustment factor between measurement site and analysis method or systematically increasing the uncertainty associated with haemoglobin concentrations obtained through non-gold standard methods. Fifth, because the elevation adjustment equation does not incorporate uncertainty, the absolute differences calculated between the WHO 2024 and WHO 2001 envelopes primarily reflect a shift of the distribution rather than additional variability, resulting in narrow UIs when comparing differences between adjustment methods. Finally, our analysis does not currently account for smoking, which has a compounding effect that can be added to the elevation adjustment for haemoglobin concentrations.¹² We intend to produce an updated set of results that incorporates smoking data and re-estimates the global burden of anaemia using haemoglobin concentrations adjusted for both elevation and smoking, in accordance with the WHO 2024 recommendations.

This study highlights the significant increases in estimated anaemia burden with the adoption of the new empirical haemoglobin elevation adjustment method by WHO in 2024. By addressing biases in previous adjustment methods, the WHO 2024 elevation adjustment method offers a more reliable and evidence-based framework for estimating anaemia burden. All future assessments of anaemia burden should adopt these new guidelines for anaemia monitoring and surveillance, including updating of historical estimates. These findings—especially that the newly enumerated cases are concentrated in low-SDI locations, young children, and women of reproductive age—underscore the clear and present need to focus efforts on prevention and treatment of anaemia in the most vulnerable populations.

Contributors

CJT, WMG, TSN, TAM, HAT, and NJK wrote the first draft of the manuscript. CJT, WMG, BCH, KH, CJLM, and NJK applied analytical methods to produce estimates. CJT, WMG, HAT, JMM, BCH, HJT, and NJK sought, catalogued, extracted, or cleaned data. CJT, WMG, and NJK designed or coded figures and tables. CJT, WMG, TSN, NP, ZAB, DG, SIH, CJLM, KMM, ON, PSS, and NJK provided data or critical feedback on data sources. CJT, WMG, BCH, KH, CJLM, and NJK developed methods or computational machinery. CJT, WMG, TSN, TAM, NGAN, AAH, NP, ZAB, DG, SIH, JMM, CJLM, KMM, ON, PSS, and NJK provided critical feedback on methods or results. CJT, WMG, TSN, TAM, HAT, NGAN, AAH, NP, ZAB, DG, SIH, JMM, CJLM, KMM, ON, PSS, and NJK drafted the work or revised it critically for important intellectual content. CJT, WMG, TSN, TAM, AAH, NP, SIH, CJLM, and NJK managed the estimation or publications process. CJLM and NJK managed the overall research enterprise. CJT, WMG, and NJK verified the underlying raw data and were responsible for the decision to submit the manuscript for publication. All other authors had access to and reviewed estimates as part of the research evaluation process.

Declaration of interests

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Data sharing

To download the data used in these analyses and corresponding results, please visit the Global Health Data Exchange at <http://ghdx.healthdata.org>.

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